Shotgun App
Vision

1. Introduction

Nowadays, private cars are one of the most used means of transport for all types of journeys: from easy travel to work or to university, to a long trip between two big cities. However, many of these journeys are made with empty seats while other people are travelling along similar routes, which contributes to more pollution and traffic issues. Shotgun is a carpooling application designed to connect drivers with available seats and passengers travelling along similar routes. The main goal of the project is to make shared travel more accessible, safe and convenient while reducing travel costs, traffic congestion, and environmental impact.

2. Positioning

2.1 Problem Statement

The problem of	There are many private cars on the road with only the driver, even though there are many other people taking the same route at the same time.
Affects	Passengers, drivers
the impact of which is	As there are many cars with only one traveller, the traffic increases, causing long queues in the road and increasing the risk of accidents. Moreover, the amount of pollution and CO2 emissions increases. At the same time, drivers don't make efficient use of their available seats and travellers spend too much money in their journeys when they could find a more affordable way to travel.
a successful solution would be	Creating a platform where passengers and drivers could connect and agree to travel together if they have to take similar routes, in order to reduce travel costs, traffic congestion, pollution, and unused seats.

2.2 Product Position Statement

For	Travellers (drivers and passengers)
Who	Travel by car in the same routes and can benefit from sharing their journeys.
The (product name)	Shotgun is a carpooling app.
That	Allows passengers to find drivers who plan to travel to their same destination, while also enabling drivers to find passengers to use their available seats in exchange for a price per seat set by the driver.
Unlike	Particular cars, taxis, Uber, public transport
Our product	Connects passengers and travellers having similar routes in a safe and friendly way and allows them to share the transport. The app shows how much CO_2 each shared ride saves, so users

	see how they contribute to improving traffic and environmental issues.

3. Stakeholder Descriptions

3.1 Stakeholder Summary

Name	Description	Responsibilities
Driver	Car owners or drivers with empty seats who offer rides along a shared route	It provides details on the route, schedules, available seats, baggage space and car model. It ensures safety during the journey. The system shows its ratings on a public profile
Passengers	People needing to travel to the same destination	Book trips, state how much baggage they bring and must respect appointment times. The system allows for the evaluation of the driver
Development team	Software engineers responsible for building, maintaining, and deploying the app's infrastructure	Ensures the architecture is scalable and maintainable over time. Monitors bugs, and continuous releases
Product Manager	Who defines the product roadmap and the alignment between business requirements and MVP features	Monitors product progress and ensures there is genuine market demand
Data protection officer	Legal compliance and privacy (GDPR) expert	It ensures that the use of data respects user privacy
Local public authority or university/company	An entity that promotes sustainable mobility among its members or employees	It defines any concession and incentives. It can follow the impact of carpooling on the public environmental statistics page

3.2 User Environment

The primary interaction typically takes place on a one-to-many basis (one driver and 1 to 6 passengers per trip).

- Initial phase: Creating a ride offer or searching for/booking a trip. Takes place while stationary, hours or days before the journey. (approx. 3 minutes)
- Transit phase: No interaction needed from the driver (hands on the wheel) or the passengers. (can last 60+ minutes)
- Post-trip phase: Rating the driver or the passengers and seeing the CO_2 saved. (less than 1 minute)

Users use the app while walking to the pickup point, often in varying weather conditions. High color contrast and large buttons are required. Drivers must not interact with complex touchscreens while driving. Critical interactions must be quick or completed before the vehicle starts moving.

We want to create a working prototype, so our goal is to build a web app. Can be used on both mobile phones and computers.

The business model is based on non-profit cost-sharing, designed to increase vehicle occupancy rates and reduce travel-related emissions. Drivers lower their travel costs by splitting expenses for fuel and tolls without generating commercial profit, while passengers benefit from an affordable transport option. Users are motivated by environmental statistics: every profile shows how much CO_2 the user has saved, and a public page shows the total for the whole platform. The meeting phase from the first version of this document (real-time vehicle location and delay notifications) was removed, because it needs a native mobile app and a maps service.

4. Product Overview

Every sprint we released a new major version of the app. The 1.0 release focused on validating the app idea with the stakeholders and users. It was frontend only, providing a basic user interface with hard coded sample data, giving early users a basic feel for the application itself. We wanted to get a reaction from the stakeholders first, before we implemented a proper backend and database to make the app fully functional.
With the 2.0 release the app became fully functional: data is stored in the database, a proper backend is implemented and the user interface is unified into a consistent, well-defined visual language.
Releases 3.0 - 5.0 all added functionality to the app itself.

4.1 Needs and Features

Need	Priority	Features	Planned Release	Delivered
Users can identify who they are travelling with before booking the ride	High	User registration, login and basic profile information	1.0	Sprint 1
Drivers are able to offer rides on the trip they are already making	High	Trip creation (specify from and to location, departure time, number of seats, price and driver contact)	1.0	Sprint 1
Passengers are able to find trips that match where and when they are going	High	Trip search and filtering by departure location, destination location, date and time	1.0	Sprint 1
Passengers can reserve seats on the trip they find	High	Seat booking without driver confirmation	1.0	Sprint 1
Information is persisted and shared between all users, so the application works with real data rather than samples	High	Users, trips and bookings are retained in a database and visible to all users.	2.0	Sprint 2
App feels intuitive and has consistent visual interface	High	Consistent color palette, fonts and general user experience across all screens	2.0	Sprint 2

Drivers and passengers can see and manage their offered/booked rides	High	Grid view of offered and reserved rides with the ability to add, edit, or remove them	2.0	Sprints 1 and 2
Passengers can cancel bookings they cannot attend	Medium	Booking cancellation from the user's bookings overview, freeing the seat for others	3.0	Sprints 1 and 2
Passengers know how many people they will be sharing the car with	Medium	Drivers specify the number of available seats; passengers see remaining seats and current participants on a trip	3.0	Sprint 3
Passengers can judge whether the driver is reliable before travelling with them	Medium	Ability to rate the driver after the ride has completed. Star and written review ratings.	4.0	Sprint 3
Drivers can rate passengers	Medium	As a driver, I can rate passengers I have been given rides to. Star and written reviews show up on their profile	4.0	Sprint 3
Users can regain account access in case of forgotten credentials	Low	Users can request a password reset link which lets them set a new password. The email is written to the server log instead of being sent.	5.0	Sprint 3
Users can verify that nobody else has access to their account	Low	Page showing all active user sessions and the ability to end them	5.0	Sprint 4
Users need a reason to choose carpooling over driving alone	Low	Gamification of the app - show leaderboard of most active drivers and passengers, provide them with in app badges and achievements	5.0	Dropped
Stakeholders need to see the environmental impact of carpooling over driving alone	High (was Low)	Drivers choose their car model with its CO_2 per km. Each ride shows its CO_2. A public page shows the number of rides shared, distance covered and CO_2 saved. Every profile shows the CO_2 the user saved.	5.0	Sprints 4 and 5
Passengers can provide their baggage information to drivers	Low	Drivers set how many small and large bags fit in the car, and passengers say how many they bring when booking. Full rides are marked in the search results.	5.0	Sprint 4

Users can see more about the people they travel with	Medium	Public profiles with a profile picture and a short bio	Not planned	Sprints 3 and 5

5. Other Product Requirements

Requirement	**Priority**	**Planned Release**	**Status**
Shotgun is a web app that works in current versions of Chrome, Firefox, Safari and Edge, on both phones and computers	High	1.0	Done
Anyone on the team or a reviewer can start the app on their own computer by following a written setup guide in under 15 minutes (Installation)	High	2.0	Done, using Docker and the README
Trip search returns results within 2 seconds (Performance)	High	2.0	Not measured
The number of bookings on a trip never exceeds its seats, even when several passengers book at the same moment (Robustness)	High	2.0	Not tested under load
A first-time user can create a trip offer or book a seat within 3 minutes, without help (Usability)	High	2.0	Not measured
Trip search, booking and cancellation are covered by automated tests that run before each release (Maintainability)	High	2.0	Done, tests run on every pull request
Passwords are stored securely, and contact details are visible only to members of the same trip (Security)	High	2.0	Passwords are hashed with bcrypt. Contact details are visible to every logged-in user on public profiles.
Drivers never need to use the app while the car is moving (Safety and usability)	High	3.0	Done
Seat prices cover only a share of fuel and toll costs, so drivers do not make a profit (Business constraint)	Medium	4.0	Not enforced
A help page explains how to offer a trip, book a seat and cancel (Documentation)	Medium	4.0	Dropped, the README and demo video explain the app

Assumptions and dependencies

- Users have a phone or computer with internet access.

- Locations come from a built-in list of European cities with a table of distances between them. The first version depended on an external maps service.
- The app does not handle payments.
- Release 1.0 uses sample data, so data, security and performance requirements apply from 2.0.